## Exercise 2: Extensively Drug-Resistant *Salmonella enterica* Serovar Typhi in Pakistan

### Introduction

Typhoid fever is a bacterial infection caused by the bacterium *Salmonella enterica* serovar Typhi. The bacteria spreads via contaminated food and water, particularly in communities with poor sanitation, and is highly contagious. Symptoms include fever, stomach pain, headache and constipation or diarrhoea. If left untreated, the infection can be fatal. Vaccination, access to clean water, and improved sanitation are effective means to prevent typhoid.

Typhoid can be treated with antibiotics, although antibiotic-resistant *Salmonella* Typhi (S. Typhi) strains have become increasingly prevalent. Historically, typhoid has been treated with ampicillin, trimethoprim-sulfamethoxazole, and chloramphenicol.[17] *S.* Typhi strains that are resistant to all these three first-line antibiotics are labelled as multidrug resistant (MDR). Resistance to fluoroquinolones has also been reported in regions where MDR infections are treated with these second-line antibiotics. Ceftriaxone, a third generation cephalosporin, and azithromycin, a macrolide, are reserved to treat multidrug-resistant infections.

In November 2016, an unusually high number of ceftriaxone-resistant cases were identified in the province of Sindh, Pakistan, primarily from the cities of Hyderabad and Karachi. The provincial public health authorities in Pakistan launched an investigation into possible sources and control measures including an emergency vaccination campaign were put in place. Around the same time, scientists at Public Health England identified a strain of typhoid with the same high levels of resistance from an individual in the United Kingdom returning from Pakistan.

The cases were resistant to five antibiotics in total: chloramphenicol, ampicillin, trimethoprim-sulfamethoxazole, fluoroquinolones, and third-generation cephalosporins. Although MDR, quinolone-resistant and sporadic ceftriaxone resistance cases had been reported before in Pakistan, this was the first time such high level of drug resistance was seen in Typhoid cases. The investigators labelled these cases as extensively drug-resistant (XDR). The doctors were left with few antibiotics available to effectively treat the infection. Scientists at the Wellcome Sanger Institute in Cambridge were approached by scientists at Aga Khan University with a request for genetic analysis of samples.[18]

### Available data

Whole-genome sequencing was carried out on 87 of the XDR *S.* Typhi strains isolated in Sindh, Pakistan, over a 6-month period between November 2016 and March 2017.

Twelve contemporaneous ceftriaxone-susceptible isolates collected from the same region were also sequenced for context. The authors found that all XDR isolates and 11 out of 12 of the contextual (ceftriaxone-sensitive) isolates belonged to the same clade (4.3.1/H58 clade). They constructed a maximum-likelihood phylogenetic tree with the XDR and contextual strains from Pakistan and previously published *S.* Typhi genomes from the same clade (4.3.1/H58 clade). We will make use of this phylogenic tree along with metadata on the country of origin, year of isolation and antibiotic resistance genetic determinants to determine the origin of these XDR *S.* Typhi cases.

### Creating a Microreact project

The phylogeny and metadata files were used to create a MicroReact project ([https://microreact.org/project/xdrtyphi](https://microreact.org/project/xdrtyphi)). You can make use of this URL to navigate to the project page or follow the instructions below (Figures 1 to 3) to create a project page from scratch.

Start by opening up a new window in Firefox and typing https://Microreact.org/ in the address bar. Click on “Upload” as show in Figure 1 to create a new project.

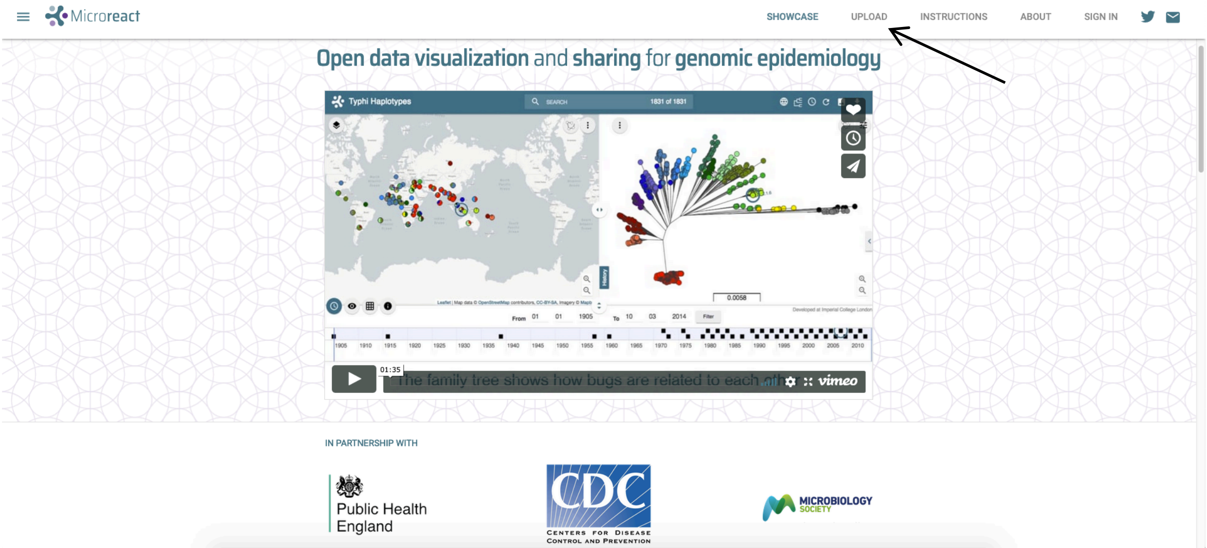



**Figure 1 Microreact home page**

Drag and drop the files Klemm2018_metadata.csv and Klemm2018_tree.nwk into the browser window.

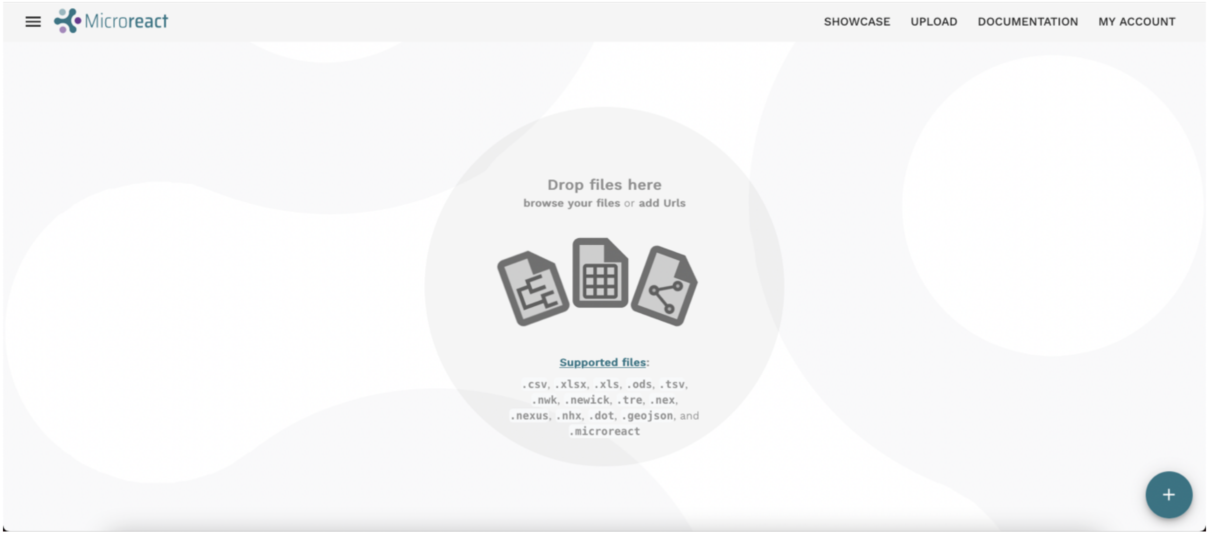



**Figure 2 Microreact upload page**

The files will start uploading automatically, once done you should see a dialogue box like the one in **Figure 3**, indicating that the files have been correctly identified as a data file (CSV or TSV format) and tree file (newick format). Click Continue.

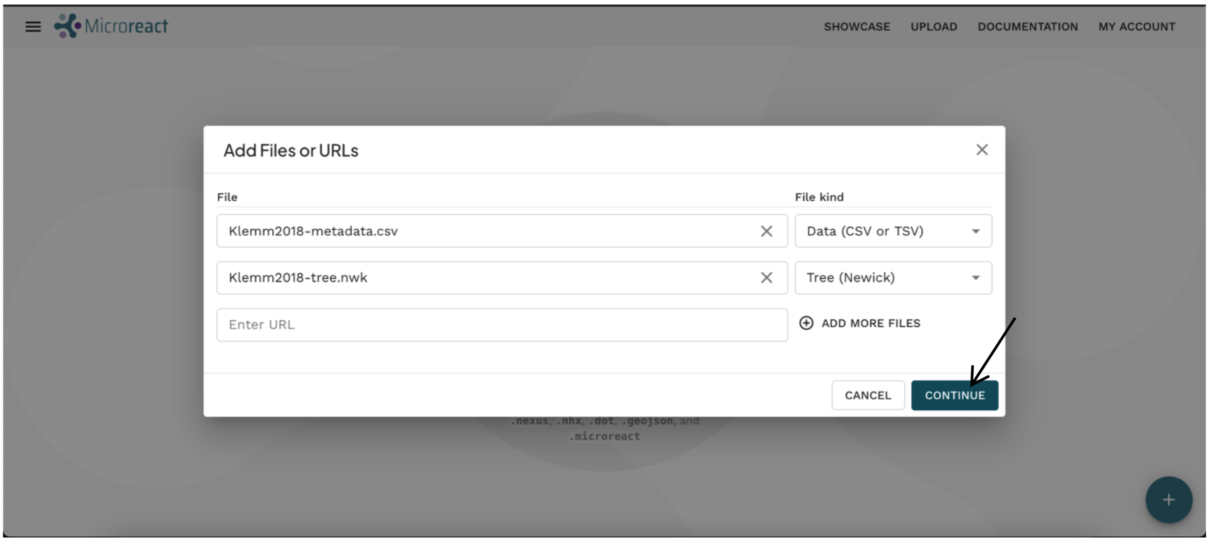



**Figure 3 Microreact uploaded files**

Once the files are loaded you should see a Microreact setup window (**Figure 4**) asking to indicate which column in the data file contains the 'labels' column – this is to ensure the data can be matched correctly to the tree file. Select the 'Id' column (arrow 1) and click Continue (arrow 2).

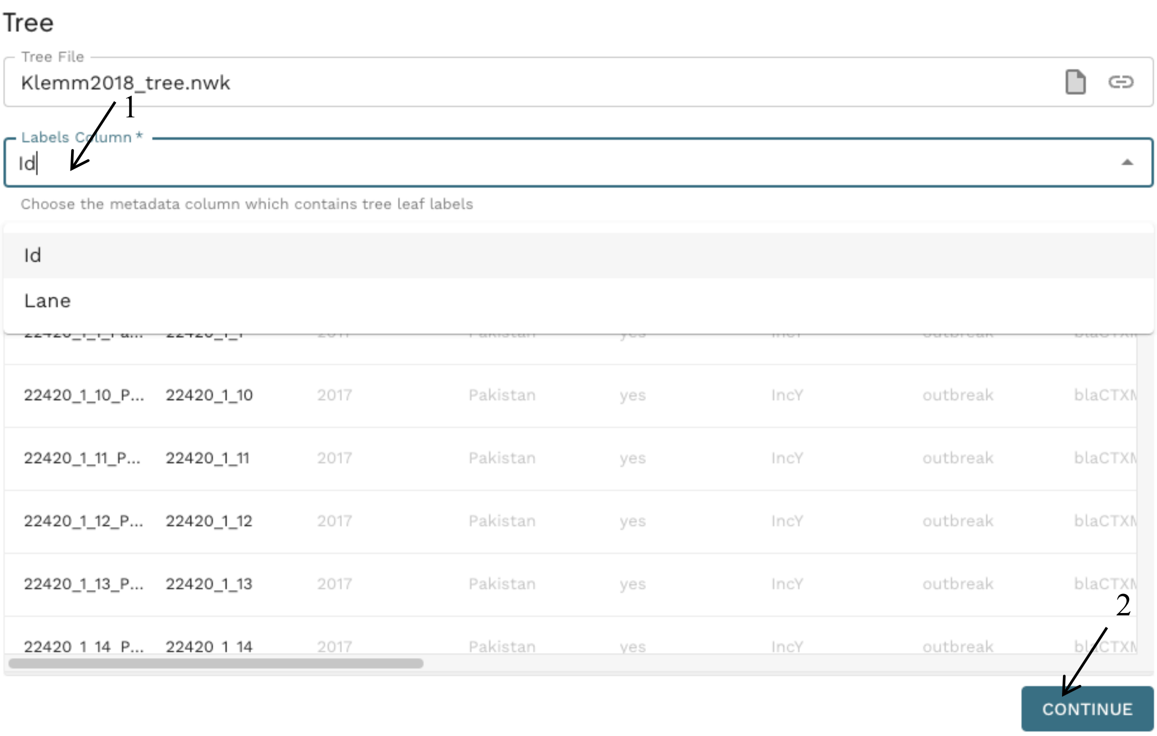



**Figure 4 Microreact setup**

You should now have a view like the one shown in Figure 5. You can use the save button (top right) to name and save the project for future use if you like.

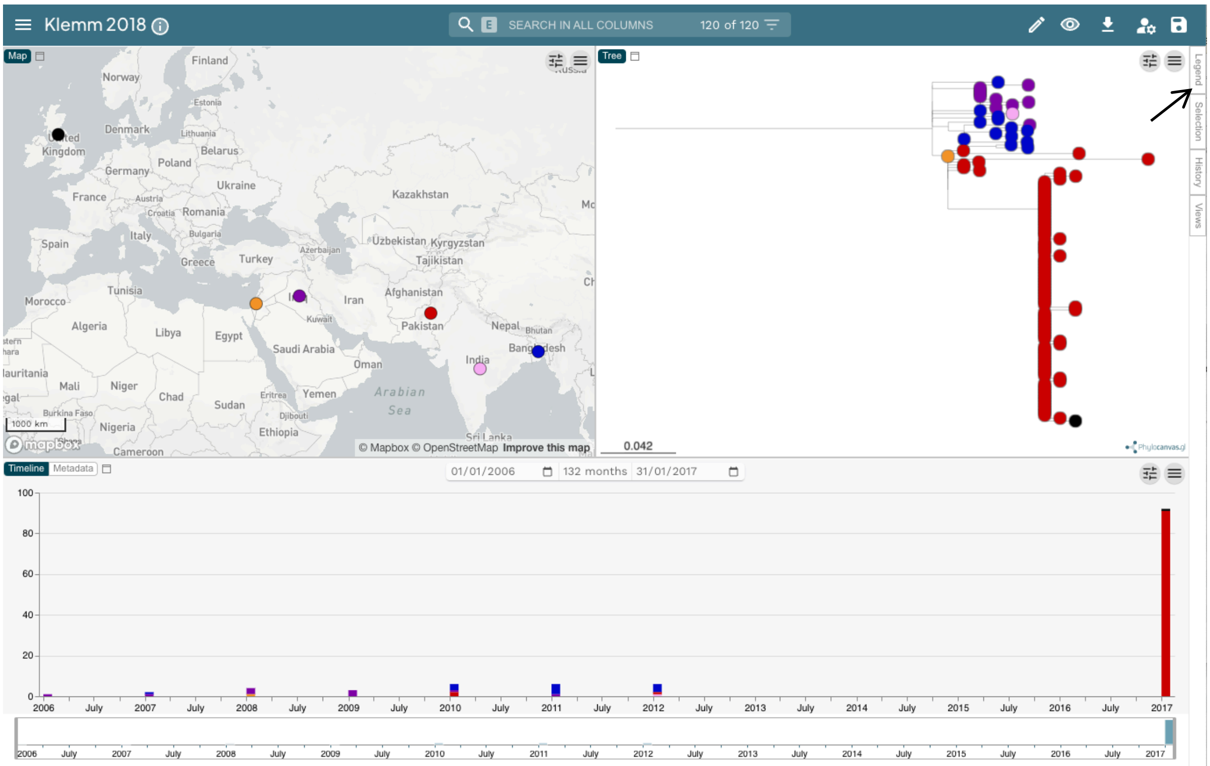



**Figure 5 Klemm2018 Microreact project**

**Identifying the origin and spread of extensively drug-resistant *S.* Typhi in Pakistan**

We will use the phylogeny and the map to identify the origin of the XDR *S.* Typhi cases in Pakistan.

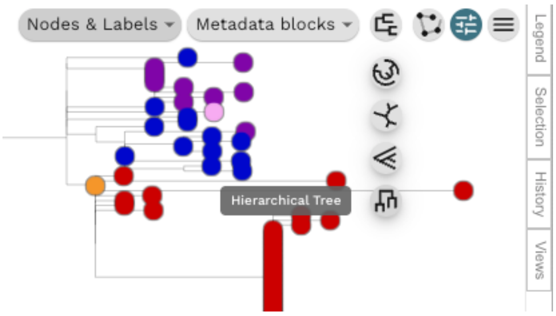
By default, the tree tips should be coloured by Country (as per Figure 5). The map view shows the colours assigned to each country, you can also click the ‘Legend’ tab (top right) in the Microreact window (arrow in Figure 5) to see a legend. It should be easy to see that most of the isolates from Pakistan (red) cluster together, separately from strains from other countries.

We can reorient the tree horizontally to make it easier to view.

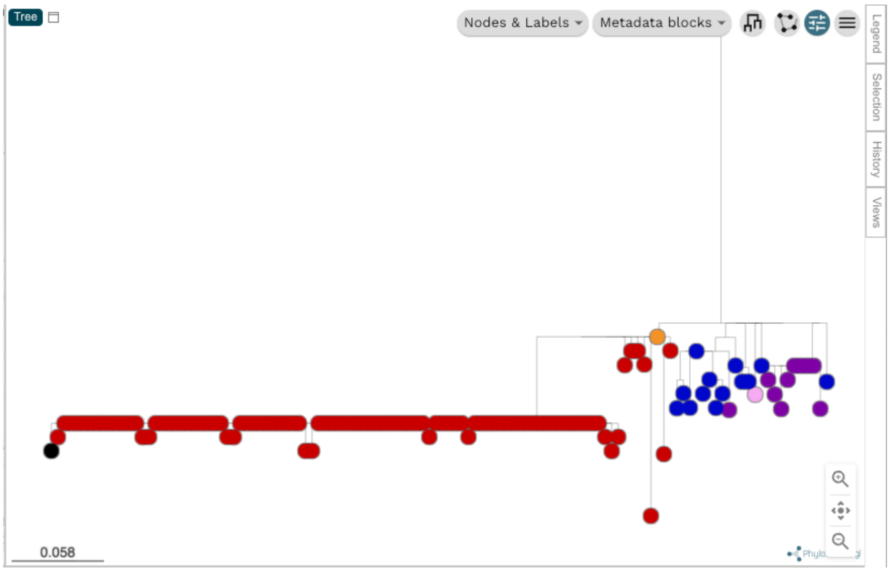
Click the control panel symbol

then click the tree-view control button

then select the ‘Hierarchical tree’ button (the last option at the bottom)



The tree should now be oriented like this:

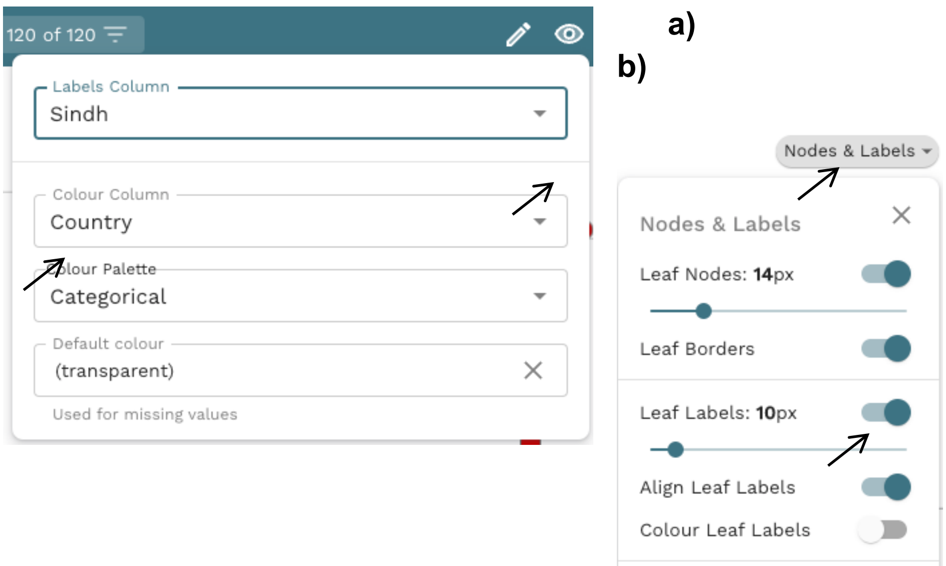



Now let’s look at the dates each strain was isolated, and which ones were epidemiologically associated with the suspected outbreak.

First, to label the outbreak strains, click the eye symbol at the top right, and then set the ‘Labels Column’ to ‘Sindh’ (Figure 6a).

Then click the ‘Nodes & Labels’ menu (above the tree) and toggle the ‘Leaf Labels’ button to switch on labelling (Figure 6b). Change the size to 10px (using the sliding scale) to make it easier to see all the labels in one view.

**Figure 6**



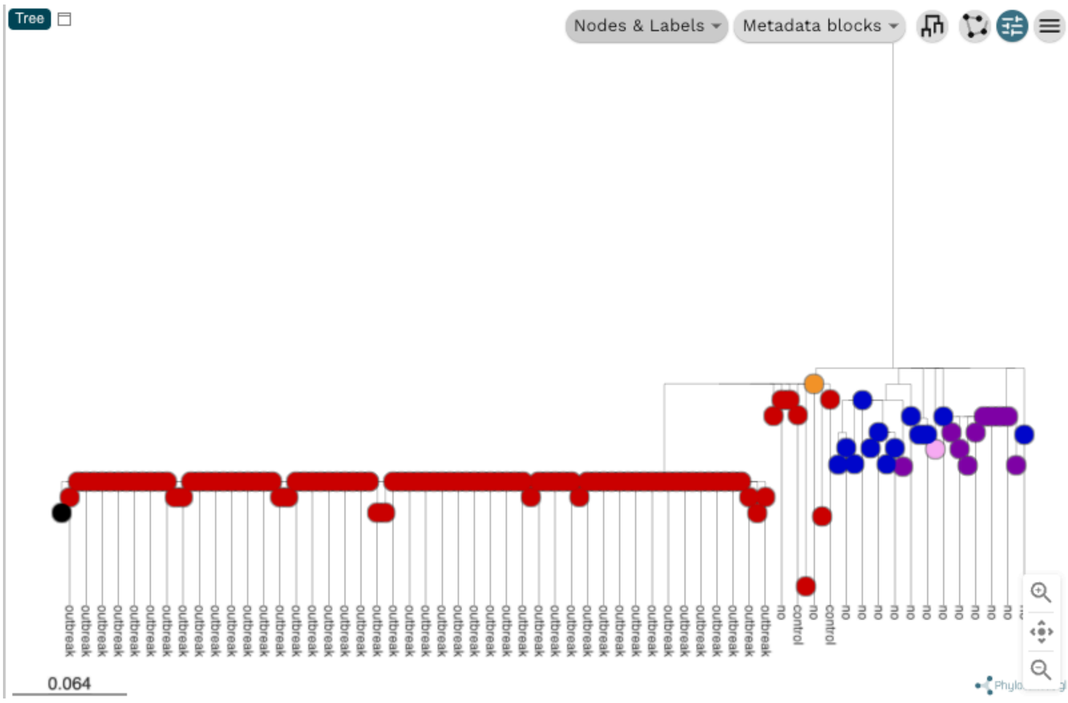
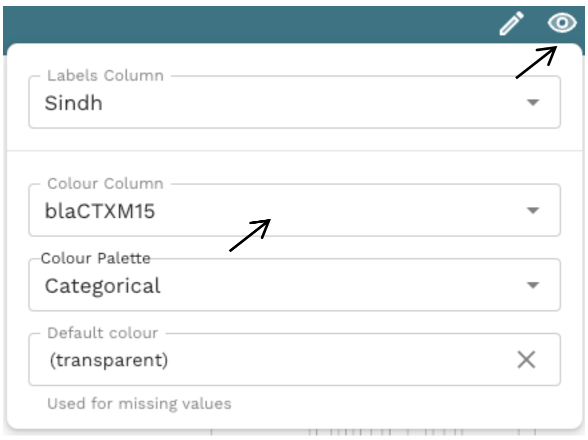



**Figure 7 Displaying the tree with hierarchical layout, country of isolation, and outbreak status.**

You should now see a view like the one in Figure 7, where it becomes clear that the isolates sequenced from the suspected outbreak in Sindh all form a tight cluster, distinct from the non-outbreak strains from Pakistan and other strains from other countries.

The authors used whole-genome sequencing to build a phylogeny of all the *S.* Typhi strains but also to investigate the genetic determinants of antibiotic resistance in this outbreak clone and in the rest of samples. You can use Microreact to view the distribution of antibiotic resistance genes against the phylogeny, to explore the genetic properties of the outbreak strain in the context of related strains, in two ways.

One option is to colour the tips of the tree by a specific AMR gene of interest. Click the eye symbol top right and change ‘Colour Column’ to a resistance gene – for example, the blaCTXM15 gene, which is responsible for the phenotype of resistance to third-generation cephalosporins such as ceftriaxone.




You should now have a view that looks like Figure 8.

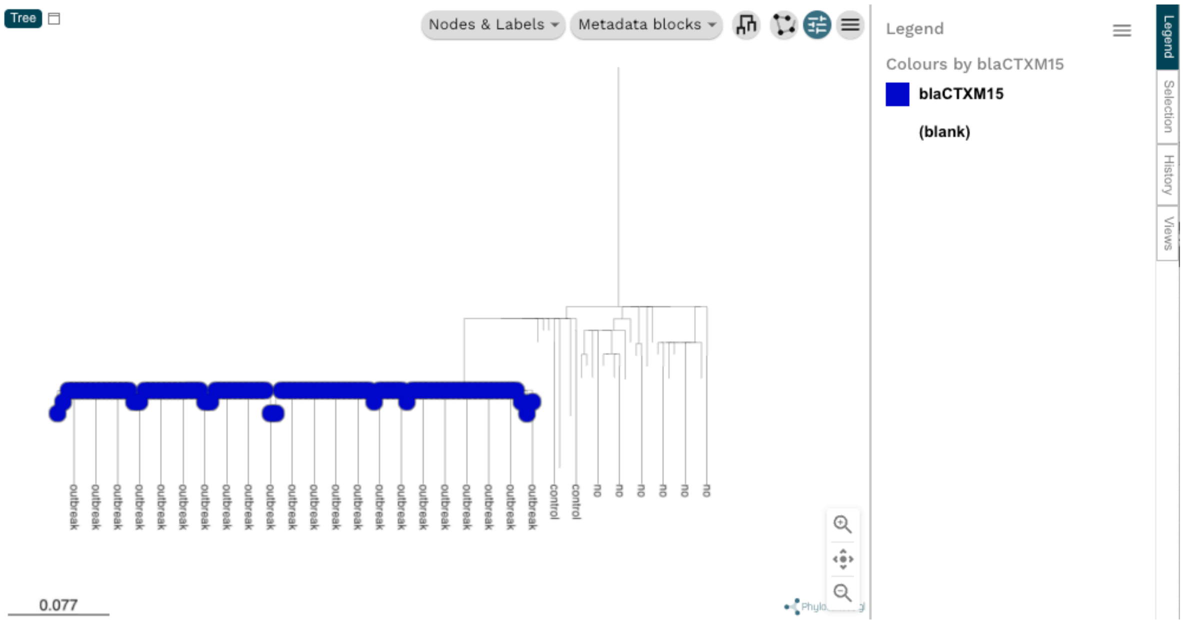
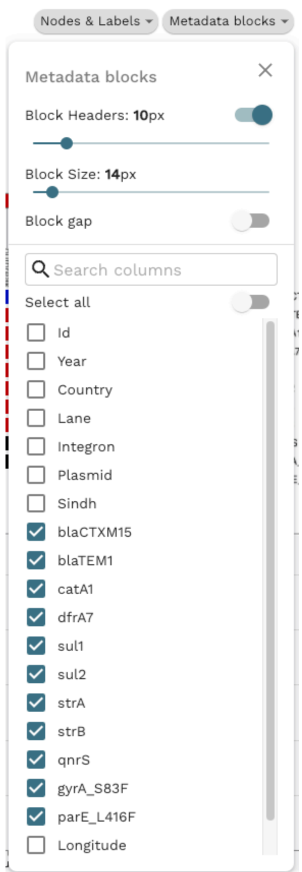



**Figure 8 Displaying the tree with a resistance determinant overlaid**

You can explore the distribution of other AMR determinants, such as the *qnrS* gene (plasmid-borne fluoroquinolone resistance), *catA1* gene (which confers resistance to chloramphenicol), *blaTEM-1* (ampicillin), *dfrA7* (trimethoprim), *sul1* and *sul2* (sulfamethoxazole), and *strA* and *strB* (streptomycin). Which ones are specific to the outbreak and which are common in other strains?

It is also possible to view multiple variables against the tree simultaneously. Click the ‘Metadata blocks’ menu button and check all the boxes that relate to resistance determinants (from blaCTXM15 down to parE_L416F).



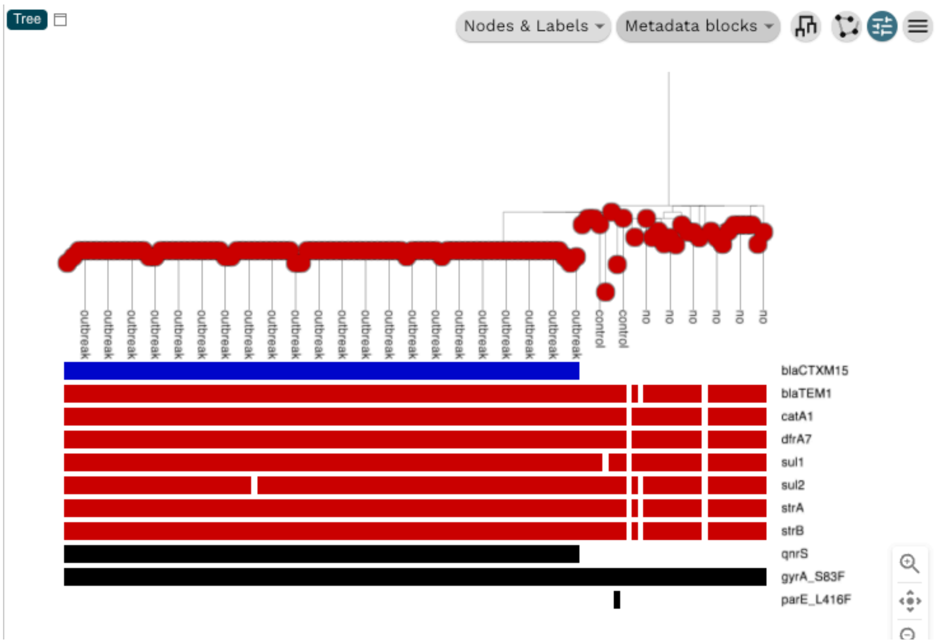




**Figure 9 AMR determinants displayed as metadata blocks.**

You will notice that almost all samples in the tree (including Sindh outbreak strains) harbour *catA1* gene (chloramphenicol resistance), *blaTEM-1* (ampicillin), *dfrA7* (trimethoprim), *sul1* and *sul2* (sulfamethoxazole), and *strA* and *strB* (streptomycin). All these genes are present in the same composite AMR transposon that is integrated into the chromosome.

The label 'Integron' under 'Label by' indicates whether samples have this AMR transposon integrated or not. The stability of this AMR cassette (i.e. it is present in almost all samples from the same phylogenetic lineage and rarely lost) is characteristic of chromosomally-integrated transposons which, once integrated into the chromosome, are inherited vertically. All samples also have the S83F mutation in *gyrA* which increases the minimum inhibitory concentration of fluoroquinoles.

The Sindh outbreak clone carries two additional AMR genes: *blaCTX-M-15* which confers resistance to ceftriaxone (a third-generation cephalosporin) and *qnrS* which mediates resistance to ciprofloxacin, both carried on the same IncY plasmid. You can see the presence of the plasmid either by adding 'Plasmid' to the list of variables shown in the 'Metadata blocks', or by setting the 'Colour Column' to 'Plasmid' (click on the eye symbol to get this menu).

Now let's consider what is the most plausible origin of the Sindh XDR outbreak clone. First, set the 'Colour Column' back to 'Country' (click on the eye symbol to get this menu), to show the country of isolation of all samples on the tree. Second, identify the genetically closest strains to the outbreak clone (circled in Figure 10). You will notice that, except for one sample isolated from Palestine in 2008, these strains were sampled in Pakistan between 2010 and 2017. The fact that the most genetically related strains to the Sindh XDR outbreak clone were also isolated in the same country earlier in time, suggest the outbreak clone derived from an endemic Pakistan clone as opposed to being imported from a different country.

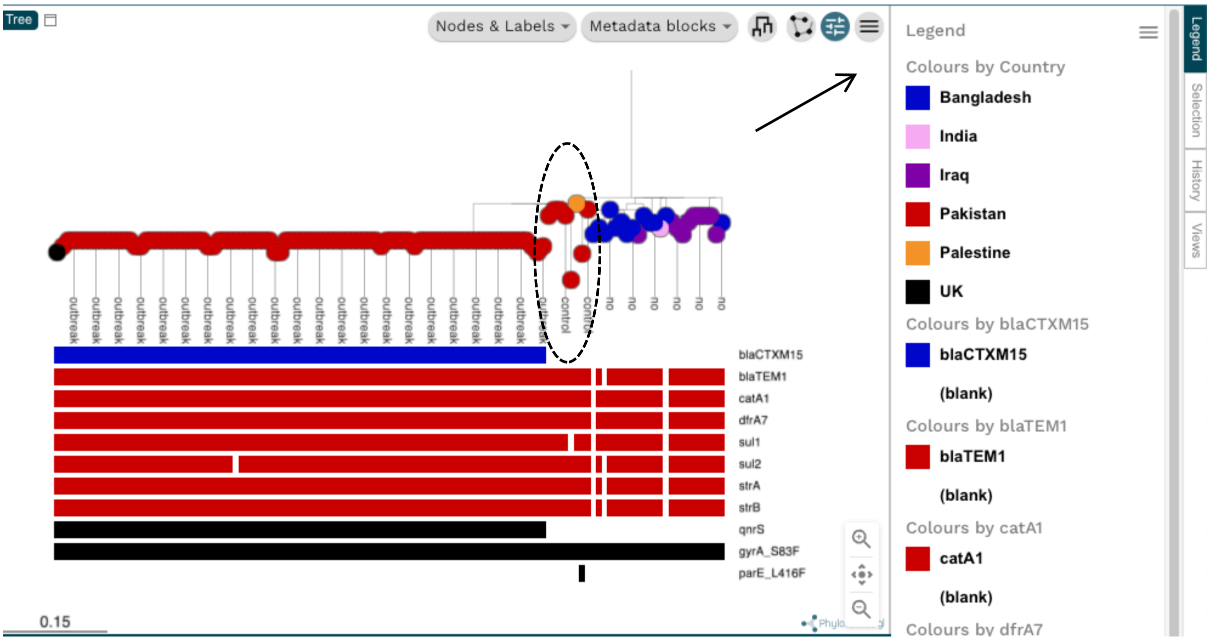


Figure 10 Identifying the origin of the Sindh XDR outbreak clone